WEEK 5 — DATA ANALYSIS & AI CHEAT SHEET

pandas + Statistics + Anomaly Detection | Data Science with ESP32 | Grade 8

1 PANDAS BASICS

```
import pandas as pd

# Load CSV file
df = pd.read_csv("dht11_data.csv")

# Explore
print(df.head())     # first 5 rows
print(df.shape)      # (rows, columns
print(df.columns)    # column names
print(df.dtypes)     # data types

# One column
temps = df["temperature"]
print(temps.head())

# Filter rows
hot = df[df["temperature"] > 35]
print(f"Hot: {len(hot)} readings")
```

2 STATISTICS

```
mean = df["temperature"].mean()
maxi = df["temperature"].max()
mini = df["temperature"].min()
std  = df["temperature"].std()

# All stats in one call
print(df["temperature"].describe())

# All columns
print(df.describe())

# Rounded output
print(f"Mean: {mean:.1f}C")
print(f"Max:  {maxi:.1f}C")
print(f"Std:  {std:.2f}")
```

3 ANOMALY DETECTION

```
# 2 standard deviations rule
mean = df["temperature"].mean()
std  = df["temperature"].std()

# Flag anomalies
df["anomaly"] = (
    df["temperature"] - mean).abs()

# Count anomalies
n = df["anomaly"].sum()
print(f"Anomalies: {n}")

# Show anomaly rows
anom = df[df["anomaly"]==True]
print(anom[["time","temperature"]])
```

4 VISUALISE ANOMALIES

```
import matplotlib.pyplot as plt

# Time series graph
plt.plot(df["temperature"],color="orange")

# Anomaly scatter (red dots)
colors = df["anomaly"].map(
    {True:"red", False:"cyan"})
plt.scatter(df.index,
    df["temperature"],c=colors,s=40)

# Boundary lines
plt.axhline(y=mean+2*std,color="oran
    linestyle="--")
plt.axhline(y=mean-2*std,color="orange",
    linestyle="--")
plt.legend(); plt.tight_layout()

# Save as image
plt.savefig("report.png",dpi=150)
plt.show()
```

5 FILTERING

```
# Greater than
hot = df[df["temperature"]>35]
# Less than
cool = df[df["temperature"]<20]
# Two conditions (use &)
both = df[(df["temperature"]>30)
          & (df["humidity"]>70)]
df2 = df.sort_values("temperature")
```

COMPLETE PIPELINE

1. Collect
Arduino sends serial data to Python

2. Store
Python saves to CSV with csv.writer()

3. Load
pd.read_csv("file.csv")

4. Explore
df.head() df.describe() df.shape

5. Analyse
mean max min std count

6. Detect
anomaly = abs(val-mean) > 2*std

7. Visualise
plt.plot() plt.scatter() axhline()

8. Report
plt.savefig("report.png", dpi=150)

INSTALL COMMAND

```
# Install once in terminal:
pip install pandas matplotlib seaborn

# Import every time:
import pandas as pd
import matplotlib.pyplot as plt
```

KEY TERMS

`DataFrame`	Table of data loaded from CSV
`Series`	One column of a DataFrame
`mean()`	Average of all values
`std()`	How spread out values are
`anomaly`	Value too far from the mean
`describe()`	All stats in one function call

GOLDEN RULES:

1. Install: pip install pandas matplotlib
2. Always start with: print(df.describe())
3. Anomaly = more than 2 std devs from mean